



D-Wave Advantage

D-Wave have launched the Advantage quantum computers with over 5,000 qubits. <http://dgit.in/oct20-61>



Googles aims carbon-free 2030

Google aims to power its data centers and offices using solely carbon-free electricity by 2030. <http://dgit.in/oct20-62>

ASUS ProArt Z490-CREATOR 10G

High-speed interconnects in a single package

With a lot of hardware coming out for the content creator segment, it was obvious that we'd start seeing motherboards designed to cater to the needs of creators. To that end, ASUS has introduced motherboards under their ProArt series. What exactly makes these boards more suited towards creators? Let's figure that out.

Board Layout

The ASUS ProArt Z490-CREATOR 10G has a simple black and gold aesthetic which is not that distant from their gaming boards. The RGB elements aren't present but the overall colour tone seems fairly similar. At the top, we see chunky heatsinks on three sides of the CPU socket and the fourth side has the RAM slots. The board uses a single 8-PIN EPS power connector and adopts a DrMOS VRM setup with 12+2 power stages. That's basically an ASP1900B VRM controller managing ONSEMI 302045 MOSFETs which are capable of handling 45A each. The RAM modules have a separate APW8723A controller.

The board doesn't have a 10G NIC, it uses an Intel I225-V 2.5G Ethernet controller and the 10G is made possible thanks to an Aquantia AQC107 add-in card which takes up an x4 PCIe slot on its own. So if you're installing the 10G card, then you might end up disabling a couple of SATA ports since those share resources with the lower x4 PCIe slot. This is a compromise that you don't have to make when you get a workstation motherboard.

There are two M.2 slots on the board. One between the first two x16 PCIe slots and one below the PCH. The first one doesn't share bandwidth while the second one does which is typical of consumer boards.



Price
₹30,500



DESIGN	86
PERFORMANCE.....	90
VALUE FOR MONEY.....	75
FEATURE	82

Being a creator board, the ASUS ProArt Z490-CREATOR 10G comes with plenty of display connectors on the rear I/O panel. There are two Thunderbolt 3 ports which support DisplayPort 1.4 with a max resolution of 7680x3840 at 60 Hz and if used for data transfer, can handle 40 Gbps. Using

the Thunderbolt port requires that you connect the DisplayPort OUT on your graphics card with the DisplayPort IN on the motherboard, of which there are two. Depending on which CPU you install and how many DisplayPort IN connectors are populated, you'll get a mix of outputs on the Thunderbolt 3 ports. An important note to remember is that the DisplayPort IN1, DisplayPort IN2, USB Type-C1 and USB Type-C2 ports should not be hot swapped while the motherboard is powered on.

BIOS and software

The BIOS on the ASUS ProArt Z490-CREATOR 10G is very similar to the PRIME series of motherboards from ASUS. All features and tabs are in their usual places so if you've been using ASUS boards, then it should be no trouble to navigate the BIOS.

On the desktop, you have the ProArt Creator hub software which offers set-

tings history for you to sort through all the older colour calibration profiles you might have used on the machine and there are also software grouping shortcuts to fire-up your frequently used software with a single-click. Other features include system vitals monitoring and fan speed control which are quite common.

Audio

On the ProArt Z490, ASUS is using SupremeFX S1220A which is based on the Realtek ALC1220 CODEC. We see a couple of audio-grade capacitors but not additional DACs seem to be present on the board. On the rear I/O, we have a 7.1 Ch audio interface along with an S/PDIF optical socket.

Bottomline

The ample amount of Thunderbolt 3 ports and high-speed connectivity are what set the ASUS ProArt Z490-CREATOR 10G apart from the usual "gaming" boards that ASUS makes. The 10G LAN alone costs around 8-10K if you were to just get a single add-in card, so a premium board giving you a 10G AIC for about 30K is actually a pretty good deal. Since things are a bit different on the rear with the DisplayPorts, we'd advise folks to read the manual properly before going ahead with this board. If you are to explore workstation boards then you'll end up spending a lot more so the ASUS ProArt Z490-CREATOR 10G is a great board for creators building a home workstation.

—Mithun Mohandas

SPECIFICATIONS

CHIPSET: Intel Z490 | MEMORY SUPPORT: 4x4800(OC) MHz upto 128GB | LAN: Intel I225-V Gigabit Ethernet + Aquantia 10G AIC | AUDIO: ALC 1220 | SATA: 6x SATA III + 2 M.2 Slots | EXPANSION: 3x PCIe X16(x16+x0+x4) | DIMENSIONS (LXWXD): 305 mm x 234 mm | WARRANTY: 3 years.

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